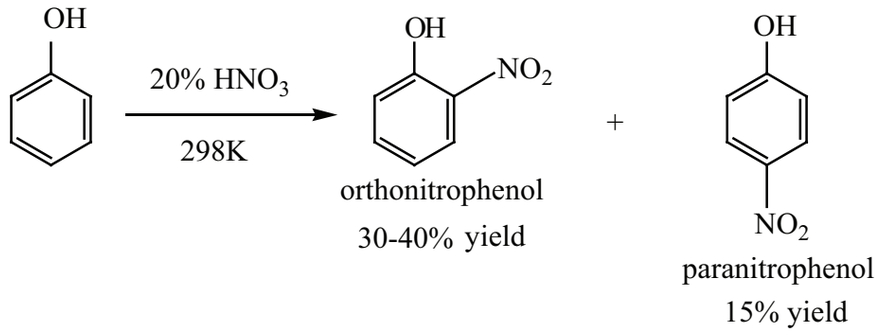
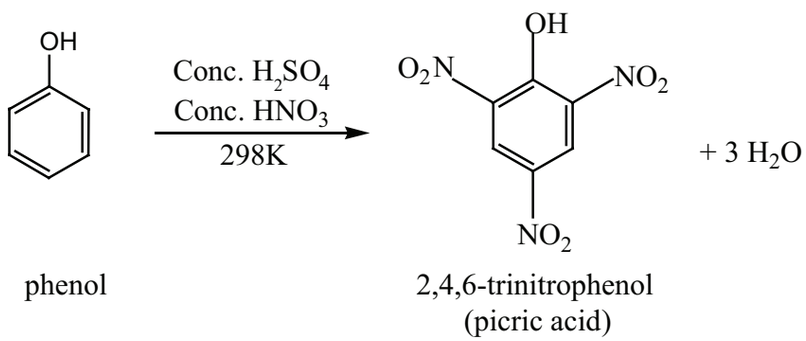
ii) Nitration:

Phenol can be nitrated using 20% nitric acid even at room temperature, a mixture of ortho and para nitro phenols are formed.



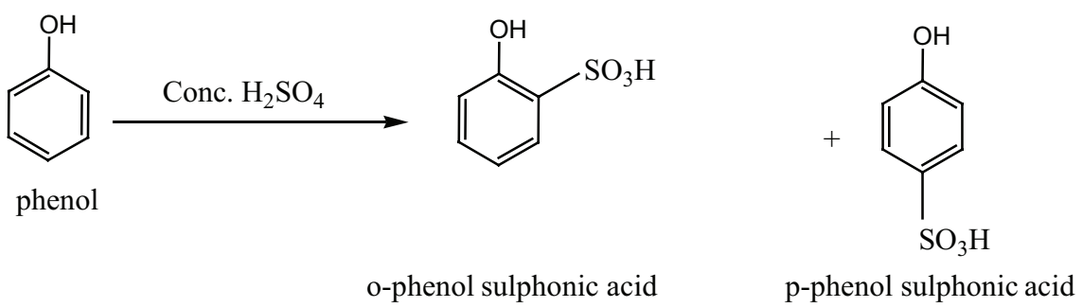
The ortho and para isomers are separated by steam distillation, as o-nitro phenol is slightly soluble in water and more volatile due to intra molecular hydrogen bonding, whereas p-nitro phenol is more soluble in water and less volatile due to intermolecular hydrogen bonding.

Nitration with Conc. HNO_3+con.H_2SO_4 gives picric acid.



iii) Sulphonation:

Phenol reacts with con.H_2SO_4 at 280K to form o-phenol sulphonic acid as the major product. When the reaction is carried out at 373K the major product is p-phenol sulphonic acid.



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